

Subject Description Form

Subject Code	ITC4216E
Subject Title	Advanced Woven Fabric Design and Structure
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	ITC3202T Woven Fabric Technology or ITC3089E Woven Fabric Design Project
Objectives	The subject provides technical knowledge required for developing advanced woven fabric designs and calculations of fabric sett for different woven designs. It explores the integration of weaving technology and woven fabric design, and thus facilitates production planning and calculating fabric price of woven fabrics.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a) learn different advanced and functional woven fabric structures and characteristics for textile product development; b) excel in fabric constructions for different advanced designs for different yarns; c) integrate the design with different weaving technologies for the production of various woven fabrics; d) create new functional fabrics structures, for different applications; e) calculate production and costing of woven fabrics;
Subject Synopsis/ Indicative Syllabus	Basics of woven fabric design: Floats in design and structure. Understanding fabric cover factor. Relation between yarn float and fabric cover. Warp and weft face design. Design repeat. Fabric sett calculation. Dobby design: Twill and Satin derivatives, Crepe, Bedford cord, Honeycomb design, Brighton honeycomb, Distorted thread effect, figured design, Pile design, Guage and Leno design. Compound design: Backed cloth, extra thread design, double cloth figuring, 3D-textile developed on 3D weaving. Jacquard design: Principles of jacquard design, Construction & Development of Jacquard Designs, Extra weft figuring, figured pile structure. Designs of various functional fabrics: Multilayer branching structures, moisture management fabrics, winter fabrics (developed based on multilayer structures with different fiber compositions), woven knitted fabric (woven fabric with knitted-like appearance), 4- way stretched fabrics(woven fabric with 4-way stretch), other functional fabrics.

Teaching/Learning Methodology	Lectures will be used to introduce principles of different types of advanced woven designs including dobby and jacquard designs, and how to calculated weaving sett for such designs. In lecture classes, students will understand machine requirements for making different designs, and learn production, and fabric price calculation. Laboratory sessions involving practical development of different fabrics learned in the theory classes. In addition, students will analyze complicated fabric designs and reproduce such fabrics. Group projects and student presentations will be the key platform for developing team spirit, self-learning ability and assist students to gain knowledge of functional and advanced fabric development. Assignments/quiz and Group projects/ presentations will be the major parts of continuous assessment components of the coursework.																																																						
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="6">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th><th>f</th></tr><tr><td>Continuous assessment</td><td>50%</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>1. Group Project and/or presentation</td><td>30%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>2. Assignment/quiz</td><td>20%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td></tr><tr><td>Examination</td><td>50%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td></tr><tr><td>Total</td><td>100%</td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table> In the continuous assessment, by group projects/presentations focus will be to develop functional fabrics and make a fabric portfolio. In this process, students will learn to work in a team for development. Assignments and quizzes will be focused to improve professional knowledge and critical and creative thinking. In the final examination, student’s theoretical and practical knowledge will be judged. Also, their understanding of woven fabric sett calculation, production calculation, and price determination will be assessed.	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)						a	b	c	d	e	f	Continuous assessment	50%							1. Group Project and/or presentation	30%	✓	✓	✓	✓	✓	✓	2. Assignment/quiz	20%	✓	✓	✓	✓	✓		Examination	50%	✓	✓	✓	✓	✓		Total	100%						
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Student Study Effort Expected	Class contact:	
	• Lecture	22 Hrs.
	• Laboratory	18 Hrs.
	• Tutorial	04 Hrs.
	Other student study effort:	
	• Project	26 Hrs.
	• Assignment	10 Hrs.
	• Self Study	25 Hrs.
	The total student study effort	105 Hrs.

Reading List and References

- Watson's Textile design and colour: elementary weaves and figured fabrics. Watson, William, Published by Forgotten Books, United Kingdom (2016)
- Watson's Advanced textile design: Compound woven structures. Watson, William, Published by Read Books, United Kingdom (2010)
- Woven Textiles: Principles, Technologies and Applications. Gandhi, K Cambridge: Elsevier Science & Technology; 2012
- Advances in 3D textiles, Xiaogang Chen (editor), Cambridge, UK: Woodhead Publishing Limited, in association with The Textile Institute; 2015
- Evolution of 3D weaving and 3D woven fabric structures, Perera, Yasith Sanura; Muwanwella, Rajapaksha, Mudiyansele Himal Widooshaka; Fernando, Philip Roshan; Fernando, Sandun Keerthichandra; Jayawardana, Thantirige Sanath Siroshana, NEW YORK: SPRINGER, Fashion and textiles, 2021-03-05, Vol.8 (1), p.1-31
- Mechanics and Calculations of Textile Machinery Gokarneshan N ; Senthil Kumar C.B ; Varadarajan B Woodhead Publishing India, 2013
- Woven fabric structure design and product planning, Hayavadana, J. (author), Boca Raton : CRC Press ; New Delhi, India : Woodhead, Publishing India Pvt Ltd ; 2015
- Structural and Mechanical Properties of Woven Fabrics Employing Peirce's Model, Boong Soo Jeon, Soo Jeon ; so Yeon Chun, Yeon Chun; Cheol Jae Hong, Jae Hong Thousand Oaks, CA: Sage Publications Textile research journal, 2003-10,
- Woolen and worsted fabric design Gilligan, E. G Cambridge: Elsevier Science & Technology ; 2004